

COMPUTER NETWORKS

Unit-IV: TCP/IP Protocol Suite

Comprehensive Study Notes

4.1 Introduction to TCP/IP Protocol Suite

Theory

The **TCP/IP Protocol Suite** is a comprehensive set of communication protocols used for the Internet and similar networks. It is named after its two most important protocols: **TCP** (Transmission Control Protocol) and **IP** (Internet Protocol).

History of TCP/IP

TCP/IP was developed in the 1970s by the US Department of Defense (DARPA). The protocol suite was designed to ensure robust and reliable communication, even under adverse conditions such as network failures or nuclear attacks. It became the standard for ARPANET in 1983 and later for the global Internet.

Key Characteristics

Characteristic	Description
Open Standard	Publicly available specifications
Protocol Suite	Collection of protocols
Scalable	Works from small to large networks
Interoperable	Different systems can communicate
Robust	Survives network failures
Flexible	Supports multiple applications

4.2 Internet Architecture Board (IAB)

Theory

The **Internet Architecture Board (IAB)** is the committee that oversees the technical and engineering development of the Internet.

IAB Structure

The IAB oversees several key organizations including the IETF (Internet Engineering Task Force), IRTF (Internet Research Task Force), and IESG (Internet Engineering Steering Group). It provides architectural oversight and final appeal in the standards process.

Key Organizations

Organization	Role
IAB	Overall architecture oversight
IETF	Develops Internet standards
IESG	Manages IETF activities
IRTF	Internet research
ISOC	Internet Society — Promotes Internet
ICANN	Domain name and IP management

RFC (Request for Comments)

RFC documents are the official technical specifications for Internet protocols.

RFC Status	Description
Proposed Standard	Under review
Draft Standard	Stable, testing
Standard	Official standard
Informational	For information
Historical	Obsolete

4.3 TCP/IP Protocol Suite Details

4.3.1 Internet Protocol (IP)

Theory: IP is responsible for **addressing** and **routing** packets across networks.

IP Characteristics

Feature	Description
Connectionless	No connection establishment
Unreliable	No guarantee of delivery
Best Effort	Makes best attempt to deliver
Datagram Service	Each packet independent

IPv4 Header Fields

Field	Size	Description
Version	4 bits	IP version (4)
IHL	4 bits	Header length
TOS	8 bits	Quality of service
Total Length	16 bits	Packet size
Identification	16 bits	Fragment identification
Flags	3 bits	Fragmentation flags
Fragment Offset	13 bits	Fragment position
TTL	8 bits	Max hops (decremented)
Protocol	8 bits	Upper layer protocol
Checksum	16 bits	Error detection
Source IP	32 bits	Source address
Destination IP	32 bits	Destination address

IP Protocol Numbers

Protocol	Number
ICMP	1
IGMP	2
TCP	6
UDP	17
OSPF	89

4.3.2 Transmission Control Protocol (TCP)

Theory: TCP provides **reliable, connection-oriented** data delivery between applications.

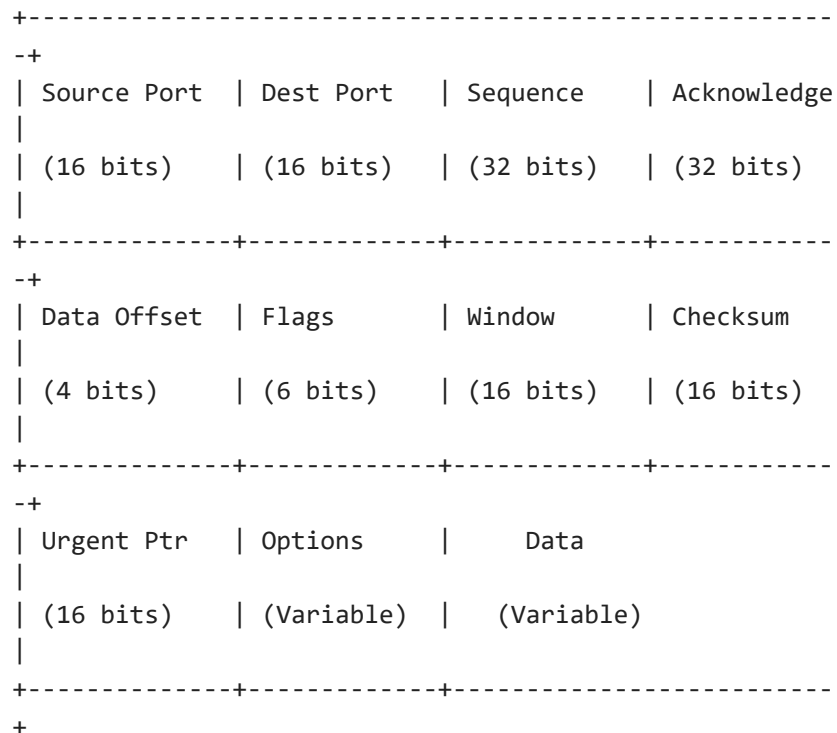
TCP Characteristics

Feature	Description
Connection-Oriented	Connection established before data
Reliable	Guaranteed delivery
Ordered	Data delivered in order
Flow Control	Prevents sender overflow
Congestion Control	Manages network congestion
Error Control	Retransmission of lost data

Three-Way Handshake

TCP uses a three-way handshake to establish a connection: (1) SYN — client sends SYN to server; (2) SYN-ACK — server responds with SYN and ACK; (3) ACK — client sends ACK, connection established. Connection termination uses a four-way handshake with FIN flags.

TCP Segment Format



TCP Flags

Flag	Name	Purpose
SYN	Synchronize	Connection establishment
ACK	Acknowledge	Data receipt confirmation
FIN	Finish	Connection termination
RST	Reset	Connection reset
PSH	Push	Immediate delivery
URG	Urgent	Urgent data

TCP States

TCP goes through several states during its lifecycle: CLOSED, LISTEN, SYN-SENT, SYN-RECEIVED, ESTABLISHED, FIN-WAIT-1, FIN-WAIT-2, CLOSE-WAIT, CLOSING, LAST-ACK, and TIME-WAIT.

TCP vs UDP Comparison

Feature	TCP	UDP
Connection	Connection-Oriented	Connectionless
Reliability	Reliable	Unreliable
Ordering	Ordered	No ordering
Flow Control	Yes	No
Congestion Control	Yes	No
Error Control	Yes	No
Header Size	20-60 bytes	8 bytes
Speed	Slower	Faster
Use Cases	Web, Email, FTP	DNS, Streaming, VoIP

4.3.3 User Datagram Protocol (UDP)

Theory: UDP provides a **connectionless, unreliable** transport service.

UDP Header Format

```

+-----+
--+
| Source Port  | Destination | Length      | Checksum
|              | Port (16 b)  | (16 bits)   | (16 bits)
|              |              |              |
+-----+
--+
```

Field	Size	Description
Source Port	16 bits	Sending application
Destination Port	16 bits	Receiving application
Length	16 bits	Datagram size
Checksum	16 bits	Error detection

4.3.4 ARP (Address Resolution Protocol)

Theory: ARP resolves **IP addresses** to **MAC addresses** on a local network.

ARP Process

The ARP process works as follows: (1) A host broadcasts an ARP request packet containing the target IP address; (2) The device with that IP responds with its MAC address (ARP reply); (3) The requesting host caches this mapping in its ARP cache for future use.

ARP Packet Format

```

+-----+
+
| Hardware | Protocol | Hardware | Protocol |
Operation|
| Type(16) | Type(16) | Addr Len | Addr Len | (16
bits)|
+-----+-----+-----+-----+-----+
+
| Sender MAC (6 bytes) | Sender IP (4 bytes) |
+-----+-----+-----+-----+-----+
| Target MAC (6 bytes) | Target IP (4 bytes) |
+-----+-----+-----+-----+-----+

```

ARP Operation Values: 1 = ARP Request, 2 = ARP Reply.

ARP Cache

ARP maintains a cache (ARP table) of recently resolved IP-to-MAC mappings to reduce broadcast traffic. Entries have a timeout period after which they are removed.

4.3.5 ICMP (Internet Control Message Protocol)

Theory: ICMP is used for **error reporting** and **diagnostic** functions.

ICMP Message Types

Type	Message	Use
0	Echo Reply	Ping response
3	Destination Unreachable	Cannot reach
4	Source Quench	Congestion
5	Redirect	Better route
8	Echo Request	Ping
11	Time Exceeded	TTL expired
12	Parameter Problem	Bad header

ICMP Header Format

+-----+			
Type	Code	Checksum	
(8 bits)	(8 bits)	(16 bits)	
+-----+			
Data (Variable)			
+-----+			

4.4 Physical and Logical Addressing

4.4.1 Physical Addressing (MAC Address)

Theory: A MAC address is a **unique hardware identifier** assigned to network interfaces.

MAC Address Format

Hexadecimal Format: AA:BB:CC:DD:EE:FF (6 bytes, 12 hex digits)

Binary Format: 10101010 10101011 10101100 11011101
11101110 11111111

MAC Address Structure

A MAC address is 48 bits (6 bytes). The first 24 bits (3 bytes) represent the OUI (Organizationally Unique Identifier) assigned to the manufacturer, and the last 24 bits are the NIC-specific identifier assigned by the manufacturer.

Types of MAC Addresses

Type	Example	Description
Unicast	AA:BB:CC:11:22:33	Single device
Multicast	01:00:5E:00:00:01	Group of devices
Broadcast	FF:FF:FF:FF:FF:FF	All devices

4.4.2 Logical Addressing (IP Address)

Theory: IP addresses are **logical addresses** that identify devices on a network.

IPv4 Address Format

Dotted Decimal Notation: 192.168.1.100

Binary Representation: 11000000 10101000 00000001
01100100

IPv4 vs IPv6

Feature	IPv4	IPv6
Address Size	32 bits	128 bits
Address Space	4.3 billion	340 undecillion
Format	Dotted Decimal	Hexadecimal
NAT	Required	Not needed
Security	Optional	Built-in
Example	192.168.1.1	2001:0db8:85a3:0000:0000:8a2e:0370:7334

4.5 Classes of IP Addressing

Theory

IP addresses are divided into **classes** to support different network sizes.

Class A Address

Range: 1.0.0.0 to 126.0.0.0
Network: First octet | Host: Last three octets
Default Subnet Mask: 255.0.0.0
Format: 0NNNNNNN HHHHHHHH HHHHHHHH HHHHHHHH

Feature	Value
First Bit	0
Network ID	0.0.0.0 — 127.0.0.0
Number of Networks	126 ($2^6 - 2$)
Hosts per Network	16,777,214 ($2^{80} - 2$)
Default Mask	255.0.0.0

Class B Address

Range: 128.0.0.0 to 191.255.0.0
Network: First two octets | Host: Last two octets
Default Subnet Mask: 255.255.0.0
Format: 10NNNNNNN NNNNNNNN HHHHHHHH HHHHHHHH

Feature	Value
First Bits	10
Network ID	128.0.0.0 — 191.255.0.0
Number of Networks	16,384 (2^{94})
Hosts per Network	65,534 ($2^{+8} - 2$)

Feature	Value
Default Mask	255.255.0.0

Class C Address

Range: 192.0.0.0 to 223.255.255.0
Network: First three octets | Host: Last octet
Default Subnet Mask: 255.255.255.0
Format: 110NNNNN NNNNNNNN NNNNNNNN HHHHHHHH

Feature	Value
First Bits	110
Network ID	192.0.0.0 — 223.255.255.0
Number of Networks	2,097,152 (2 ⁺⁹)
Hosts per Network	254 (2 ⁸ – 2)
Default Mask	255.255.255.0

Class D Address (Multicast)

Range: 224.0.0.0 to 239.255.255.255
Purpose: Multicast groups
Format: 1110XXXX XXXXXXXXXXXX XXXXXXXXXXXX XXXXXXXXXXXX

Class E Address (Experimental)

Range: 240.0.0.0 to 255.255.255.255
Purpose: Reserved/Experimental
Format: 1111XXXX XXXXXXXXXXXX XXXXXXXXXXXX XXXXXXXXXXXX

Address Class Summary

Class	Start Bits	Address Range	Network	Host	Mask
A	0	1–126	1 octet	3 octet	255.0.0.0

Class	Start Bits	Address Range	Network	Host	Mask
B	10	128–191	2 octet	2 octet	255.255.0.0
C	110	192–223	3 octet	1 octet	255.255.255.0
D	1110	224–239	Multicast	—	—
E	1111	240–255	Experimental	—	—

4.6 Special IP Addresses

Theory

Certain IP addresses are reserved for **special purposes** and cannot be used for general hosts.

1. Network ID

Address: All host bits are 0

Example: 192.168.1.0

Purpose: Identifies the network (cannot be assigned to a host)

2. Broadcast Address

Address: All host bits are 1

Example: 192.168.1.255

Purpose: Send to all hosts on network

Type	Address	Purpose
Directed Broadcast	x.x.x.255	All hosts on network
Limited Broadcast	255.255.255.255	All hosts on local network

3. Loopback Address

Range: 127.0.0.0 to 127.255.255.255

Common: 127.0.0.1 (localhost)

Purpose: Testing network software (cannot be used on networks)

4. Private Addresses

Class A Private: 10.0.0.0 to 10.255.255.255 (Mask: 255.0.0.0)

Class B Private: 172.16.0.0 to 172.31.255.255 (Mask:

255.255.0.0)

Class C Private: 192.168.0.0 to 192.168.255.255 (Mask:
255.255.255.0)

5. APIPA (Automatic Private IP Addressing)

Range: 169.254.0.0 to 169.254.255.255

Purpose: Automatic addressing when DHCP fails

Mask: 255.255.0.0

Special Addresses Summary

Type	Range	Purpose
Network ID	x.x.x.0	Identify network
Broadcast	x.x.x.255	Send to all hosts
Loopback	127.x.x.x	Local testing
Private A	10.x.x.x	Internal networks
Private B	172.16–31.x.x	Internal networks
Private C	192.168.x.x	Internal networks
APIPA	169.254.x.x	Automatic addressing

4.7 Classful Addressing

Theory

Classful Addressing is the original IP addressing scheme where addresses are divided into fixed classes (A, B, C, D, E) based on the leading bits.

Limitations of Classful Addressing

1. **Waste of IP Addresses** — Large address blocks assigned regardless of actual need
2. **Inflexible** — Fixed sizes make it difficult to match network sizes
3. **Inefficient** — Many addresses remain unused

4.8 Subnetting

Theory

Subnetting is the process of dividing a network into smaller sub-networks (subnets).

Why Subnetting?

Subnetting improves IP address utilization, reduces network traffic, simplifies network management, improves security by isolating network segments, and contains broadcast traffic within smaller domains.

Subnetting Steps

1. Determine the number of subnets required
2. Calculate the number of bits to borrow from the host portion
3. Determine the new subnet mask
4. Calculate the valid subnets, host ranges, and broadcast addresses

Subnetting Example

Problem: Subnet 192.168.1.0/24 into 4 subnets.

Step 1: Need 4 subnets → Need 2 bits ($2^2 = 4$)

Step 2: New subnet mask = /24 + 2 = /26 (255.255.255.192)

Step 3: Calculate subnets:

Subnet	Network Address	Host Range	Broadcast
1	192.168.1.0	192.168.1.1 – 192.168.1.62	192.168.1.63

Subnet	Network Address	Host Range	Broadcast
2	192.168.1.64	192.168.1.65 – 192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193 – 192.168.1.254	192.168.1.255

Subnet Mask Quick Reference

Prefix	Subnet Mask	Subnets	Hosts
/24	255.255.255.0	1	254
/25	255.255.255.128	2	126
/26	255.255.255.192	4	62
/27	255.255.255.224	8	30
/28	255.255.255.240	16	14
/29	255.255.255.248	32	6
/30	255.255.255.252	64	2
/31	255.255.255.254	128	1

Subnetting Formula

Calculation	Formula
Number of Subnets	2^n (n = borrowed bits)
Hosts per Subnet	$2^h - 2$ (h = host bits)
Subnet Mask	32 – (Network Bits + Borrowed Bits)

4.9 Supernetting

Theory

Supernetting (or **CIDR**) combines multiple smaller networks into a larger one.

Why Supernetting?

1. **Reduces routing table size**
2. **More efficient addressing**
3. **Simplifies network management**

Supernetting Example

Problem: Supernet 4 networks: 192.168.0.0/24, 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

4 networks = 2^2 → Need 2 less bits
New prefix = /24 - 2 = /22
Supernet = 192.168.0.0/22
Range: 192.168.0.0 - 192.168.3.255

4.10 Classless Addressing (CIDR)

Theory

CIDR (Classless Inter-Domain Routing) eliminates the classful boundaries and uses variable-length prefixes.

CIDR Notation

Format: IP Address/Prefix Length

Example: 192.168.1.0/24

CIDR Example

CIDR allows networks to be of any size (not restricted to /8, /16, /24). For example, a network requiring only 50 hosts can be assigned a /26 subnet, avoiding the waste inherent in classful addressing.

CIDR Benefits

Benefit	Description
Efficient	No address waste
Flexible	Any prefix length
Scalable	Supports Internet growth
Aggregation	Smaller routing tables

4.11 TCP/IP Service Protocols

4.11.1 FTP (File Transfer Protocol)

Theory: FTP is used for transferring files between computers on a network. It uses ports 20 (data) and 21 (control).

FTP Features

Feature	Description
Ports	20 (Data), 21 (Control)
Authentication	Username/Password
File Types	ASCII, Binary
Modes	Active, Passive

FTP Commands

Command	Purpose
USER	Login username
PASS	Password
LIST	List files
RETR	Retrieve file
STOR	Store file
DELE	Delete file
QUIT	End session

4.11.2 SMTP (Simple Mail Transfer Protocol)

Theory: SMTP is used for sending email messages between servers on port 25.

SMTP Features

Feature	Description
Port	25
Purpose	Sending email
Authentication	Optional
Format	Text-based

SMTP Commands

Command	Purpose
HELO/EHLO	Identify client
MAIL FROM	Sender address
RCPT TO	Recipient address
DATA	Message content
QUIT	End session

Email Delivery Process

Email delivery involves: sender → MUA (Mail User Agent) → MSA (Mail Submission Agent) → MTA (Mail Transfer Agent) using SMTP → destination MTA → MDA (Mail Delivery Agent) → recipient's mailbox → retrieved via POP3/IMAP.

4.11.3 TELNET (Terminal Network)

Theory: TELNET provides remote login access to computers on port 23.

TELNET Features

Feature	Description
Port	23
Purpose	Remote access
Authentication	Username/Password

Feature	Description
Security	No encryption

TELNET Commands

Command	Purpose
Open	Connect to host
Close	End connection
Display	Display options
Quit	Exit TELNET

Security Concerns

TELNET has significant security issues: no encryption, passwords sent in clear text, and vulnerability to man-in-the-middle attacks. It has largely been replaced by SSH (Secure Shell, port 22) for secure remote access.

4.11.4 DNS (Domain Name System)

Theory: DNS translates human-readable domain names to IP addresses.

DNS Hierarchy

The DNS hierarchy consists of: Root Domain (.) → Top-Level Domains (.com, .org, .in) → Second-Level Domains (google, example) → Subdomains (www, mail).

DNS Record Types

Record	Purpose	Example
A	IPv4 Address	192.168.1.1
AAAA	IPv6 Address	2001:db8::1
CNAME	Canonical Name	www → website
MX	Mail Exchange	mail.google.com
NS	Name Server	ns1.google.com
TXT	Text Record	Verification

DNS Resolution Process

DNS resolution works through recursive or iterative queries: client → local DNS resolver → root server → TLD server → authoritative name server → IP address returned to client.

DNS Ports

Protocol	Port	Purpose
DNS (UDP)	53	Standard queries
DNS (TCP)	53	Zone transfers, large responses

4.12 Complete Summary

TCP/IP Protocol Suite Summary

Layer	Protocols	Function
Application	HTTP, FTP, SMTP, DNS, TELNET	User services
Transport	TCP, UDP	End-to-end delivery
Internet	IP, ARP, ICMP	Addressing/Routing
Network Access	Ethernet, WiFi	Physical transmission

IP Addressing Summary

Class	Range	Mask	Network	Host
A	1–126	/8	8 bits	24 bits
B	128–191	/16	16 bits	16 bits
C	192–223	/24	24 bits	8 bits
D	224–239	—	Multicast	—
E	240–255	—	Experimental	—

Special Addresses Summary

Type	Address	Purpose
Network ID	x.x.x.0	Identify network
Broadcast	x.x.x.255	All hosts
Loopback	127.0.0.1	Local test
Private A	10.x.x.x	Internal
Private B	172.16–31.x.x	Internal

Type	Address	Purpose
Private C	192.168.x.x	Internal

Protocol Port Numbers

Protocol	Port	Purpose
HTTP	80	Web browsing
HTTPS	443	Secure web
FTP Control	21	File transfer
FTP Data	20	File transfer
SMTP	25	Email sending
DNS	53	Name resolution
TELNET	23	Remote access
SSH	22	Secure remote access
POP3	110	Email receiving

4.13 Previous Year Questions

Short Questions (2 marks)

1. What is IP address?
2. Differentiate between TCP and UDP.
3. What is the purpose of ARP?
4. Define subnetting.
5. What is CIDR?
6. What is the role of DNS?
7. List the classes of IP addresses.
8. What is a private IP address?

Medium Questions (5 marks)

1. Explain the TCP/IP protocol suite architecture.
2. What is IP addressing? Explain the different classes.
3. Explain subnetting with an example.
4. Compare IPv4 and IPv6.
5. Explain ARP and its working.

Long Questions (12 marks)

1. (a) Explain the TCP/IP protocol suite with all layers.
2. (a) What is IP addressing? Explain classful addressing.
3. (a) Explain the working of TCP with three-way handshake.
4. (a) Explain the following protocols: FTP, SMTP, TELNET, DNS.

Quick Reference Card

TCP/IP Layers & Protocols

Application: HTTP, FTP, SMTP, DNS, TELNET
Transport: TCP, UDP
Internet: IP, ARP, ICMP
Network Access: Ethernet, WiFi

IP Classes Memory Aid

Class A: 1-126 (Large networks)
Class B: 128-191 (Medium networks)
Class C: 192-223 (Small networks)
Class D: 224-239 (Multicast)
Class E: 240-255 (Experimental)

Special Addresses

Network ID: x.x.x.0
Broadcast: x.x.x.255
Loopback: 127.x.x.x
Private A: 10.x.x.x
Private B: 172.16-31.x.x
Private C: 192.168.x.x
APIPA: 169.254.x.x

Key Protocols & Ports

HTTP	-> 80	HTTPS	-> 443
FTP	-> 21	DNS	-> 53
SMTP	-> 25	TELNET	-> 23
POP3	-> 110	SSH	-> 22